EXPERIMENT : 13

AIM:

Demonstrate an experiment to handle missing data and inappropriate data in a Data set using Python Pandas Library for Data Preprocessing.

DESCRIPTION:

Understand the procedure to handle missing and inappropriate data

CODE:

```
import numpy as np
import pandas as pd

# Read dataset
df = pd.read_csv('/content/hotel1.csv')
# Display dataset
print(df)

# Check duplicate rows
print(df.duplicated())

# Display information
df.info()

# Remove duplicate rows
df.drop_duplicates(inplace=True)

# Check number of rows
print(len(df))

# Reset index
df.reset_index(drop=True, inplace=True)

print(df)

# Drop unwanted column if it exists
```

```
df.drop(['Age_Group.1'], axis=1, inplace=True, errors='ignore')

# Replace negative values with NaN
df.loc[df['CustomerID'] < 0, 'CustomerID'] = np.nan
df.loc[df['Bill'] < 0, 'Bill'] = np.nan
df.loc[df['EstimatedSalary'] < 0, 'EstimatedSalary'] = np.nan

# Handle invalid number of persons
df.loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20), 'NoOfPax'] = np.nan

# Check unique values
print(df['Age_Group'].unique())
print(df['Hotel'].unique())
print(df['FoodPreference'].unique())

# Correct hotel spelling
df['Hotel'] = df['Hotel'].replace('Ibys', 'Ibis')

# Standardize food preferences
df['FoodPreference'] = df['FoodPreference'].replace(
    ['Vegetarian', 'veg'], 'Veg'
)

df['FoodPreference'] = df['FoodPreference'].replace(
    ['non-Veg'], 'Non-Veg'
)

# Handle invalid ratings
df.loc[(df['Rating'] < 1) | (df['Rating'] > 5), 'Rating'] = np.nan

# Fill missing values
df['EstimatedSalary'] = df['EstimatedSalary'].fillna(
    round(df['EstimatedSalary'].mean())
)

df['NoOfPax'] = df['NoOfPax'].fillna(
    round(df['NoOfPax'].median())
)

df['Rating'] = df['Rating'].fillna(
    round(df['Rating'].median())
)
```

```
df['Bill'] = df['Bill'].fillna(
    round(df['Bill'].mean())
)

# Display final cleaned dataset
print(df)

# Check missing values
print(df.isnull().sum())

# Display final information
df.info()
```

OUTPUT:

```
   CustomerID Age_Group  Rating       Hotel FoodPreference  Bill  NoOfPax  \
0           1     20-25       4        Ibis           Veg  1300        2
1           2     30-35       5  Lemon Tree       Non-Veg  2000        3
2           3     25-30       3         Taj           Veg  2500        2
3           4     35-40       4    Radisson       Non-Veg  3200        4
4           5     20-25       2        Ibis           Veg   900        1
5           6     40-45       5         Taj       Non-Veg  4500        5
6           7     25-30       4  Lemon Tree           Veg  1800        2
7           8     30-35       3    Radisson       Non-Veg  2800        3
8           9     20-25       4        Ibis       Non-Veg  1500        2
9          10     45-50       5         Taj           Veg  4000        4
10         11     25-30       3  Lemon Tree           Veg  1600        2
11         12     35-40       4    Radisson       Non-Veg  3100        3
12         13     30-35       5        Ibis           Veg  1900        2
13         14     20-25       3  Lemon Tree       Non-Veg  1400        2
14         15     40-45       4         Taj           Veg  3800        4

    EstimatedSalary
0             40000
1             59000
2             52000
3             70000
4             30000
5             85000
6             48000
7             61000
```

```
8            36000
9            90000
10           45000
11           68000
12           56000
13           34000
14           82000
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
dtype: bool
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       15 non-null     int64
 1   Age_Group        15 non-null     object
 2   Rating           15 non-null     int64
 3   Hotel            15 non-null     object
 4   FoodPreference   15 non-null     object
 5   Bill             15 non-null     int64
 6   NoOfPax          15 non-null     int64
 7   EstimatedSalary  15 non-null     int64
dtypes: int64(5), object(3)
memory usage: 1.1+ KB
15
    CustomerID Age_Group  Rating       Hotel FoodPreference  Bill  NoOfPax  \
0            1     20-25       4        Ibis            Veg  1300        2
1            2     30-35       5  Lemon Tree        Non-Veg  2000        3
2            3     25-30       3         Taj            Veg  2500        2
3            4     35-40       4    Radisson        Non-Veg  3200        4
```

```
4           5    20-25      2        Ibis           Veg   900        1
5           6    40-45      5         Taj        Non-Veg  4500        5
6           7    25-30      4  Lemon Tree           Veg  1800        2
7           8    30-35      3    Radisson        Non-Veg  2800        3
8           9    20-25      4        Ibis        Non-Veg  1500        2
9          10    45-50      5         Taj           Veg  4000        4
10         11    25-30      3  Lemon Tree           Veg  1600        2
11         12    35-40      4    Radisson        Non-Veg  3100        3
12         13    30-35      5        Ibis           Veg  1900        2
13         14    20-25      3  Lemon Tree        Non-Veg  1400        2
14         15    40-45      4         Taj           Veg  3800        4

    EstimatedSalary
0             40000
1             59000
2             52000
3             70000
4             30000
5             85000
6             48000
7             61000
8             36000
9             90000
10            45000
11            68000
12            56000
13            34000
14            82000
['20-25' '30-35' '25-30' '35-40' '40-45' '45-50']
['Ibis' 'Lemon Tree' 'Taj' 'Radisson']
['Veg' 'Non-Veg']
    CustomerID Age_Group  Rating       Hotel FoodPreference    Bill  NoOfPax  \
0          1.0     20-25     4.0        Ibis           Veg  1300.0      2.0
1          2.0     30-35     5.0  Lemon Tree        Non-Veg  2000.0      3.0
2          3.0     25-30     3.0         Taj           Veg  2500.0      2.0
3          4.0     35-40     4.0    Radisson        Non-Veg  3200.0      4.0
4          5.0     20-25     2.0        Ibis           Veg   900.0      1.0
5          6.0     40-45     5.0         Taj        Non-Veg  4500.0      5.0
6          7.0     25-30     4.0  Lemon Tree           Veg  1800.0      2.0
7          8.0     30-35     3.0    Radisson        Non-Veg  2800.0      3.0
8          9.0     20-25     4.0        Ibis        Non-Veg  1500.0      2.0
9         10.0     45-50     5.0         Taj           Veg  4000.0      4.0
10        11.0     25-30     3.0  Lemon Tree           Veg  1600.0      2.0
11        12.0     35-40     4.0    Radisson        Non-Veg  3100.0      3.0
```

```
12        13.0     30-35     5.0        Ibis           Veg  1900.0      2.0
13        14.0     20-25     3.0  Lemon Tree       Non-Veg  1400.0      2.0
14        15.0     40-45     4.0         Taj           Veg  3800.0      4.0

    EstimatedSalary
0           40000.0
1           59000.0
2           52000.0
3           70000.0
4           30000.0
5           85000.0
6           48000.0
7           61000.0
8           36000.0
9           90000.0
10          45000.0
11          68000.0
12          56000.0
13          34000.0
14          82000.0
CustomerID          0
Age_Group           0
Rating              0
Hotel               0
FoodPreference      0
Bill                0
NoOfPax             0
EstimatedSalary     0
dtype: int64
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       15 non-null     float64
 1   Age_Group        15 non-null     object
 2   Rating           15 non-null     float64
 3   Hotel            15 non-null     object
 4   FoodPreference   15 non-null     object
 5   Bill             15 non-null     float64
 6   NoOfPax          15 non-null     float64
 7   EstimatedSalary  15 non-null     float64
dtypes: float64(5), object(3)
memory usage: 1.1+ KB
```

RESULT:

Missing values and inappropriate data in the given dataset were successfully identified and handled. The dataset was cleaned using suitable techniques, improving data quality and ensuring consistency for further analysis.